

E-Commerce Sales Analytics using SQL

Prepared By:

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Course:

Data Analytics Project

Tools Used:

- SQL
- MySQL Workbench
- Power BI

Project Type:

SQL Data Analysis Project

Year:

2026

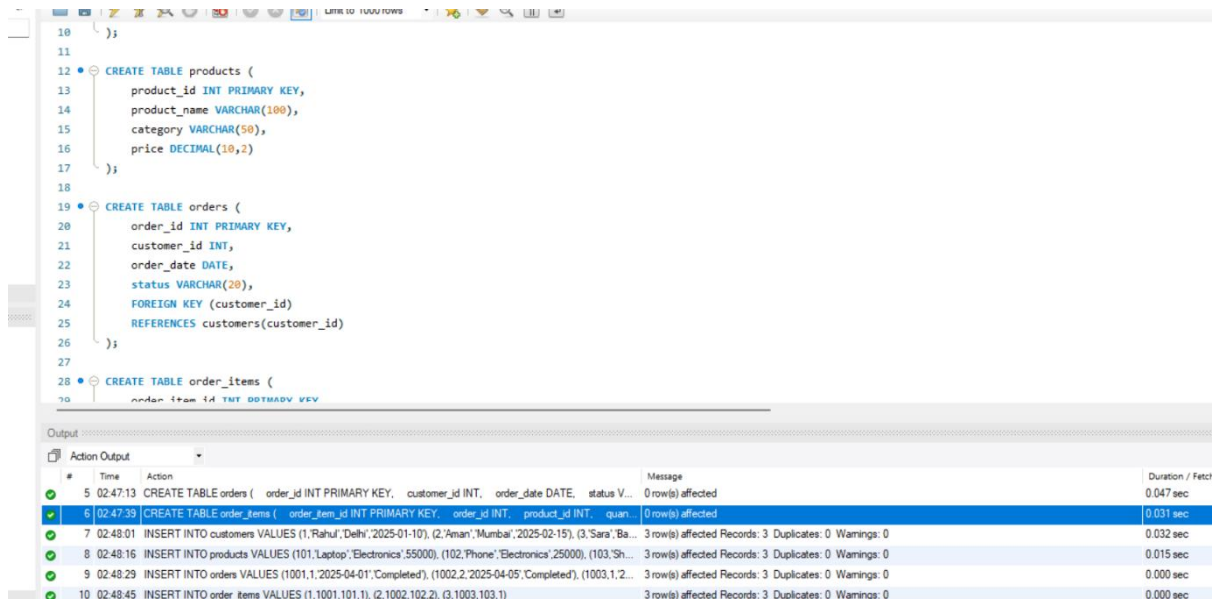
OBJECTIVE:-

The objective of this project is to analyze sales data, customer behavior, and product performance using SQL queries and data analytics techniques.

This project helps in understanding:

- Revenue analysis
- Top selling products
- Customer insights
- SQL joins and window functions
- Business decision making

SQL CREATE TABLE Queries:-



The screenshot displays a SQL IDE with three CREATE TABLE queries and their execution results in the Output window.

```
10 );
11
12 CREATE TABLE products (
13     product_id INT PRIMARY KEY,
14     product_name VARCHAR(100),
15     category VARCHAR(50),
16     price DECIMAL(10,2)
17 );
18
19 CREATE TABLE orders (
20     order_id INT PRIMARY KEY,
21     customer_id INT,
22     order_date DATE,
23     status VARCHAR(20),
24     FOREIGN KEY (customer_id)
25     REFERENCES customers(customer_id)
26 );
27
28 CREATE TABLE order_items (
29     order_item_id INT PRIMARY KEY,
30     order_id INT,
31     product_id INT,
32     quantity INT
33 );
```

The Output window shows the following results:

#	Time	Action	Message	Duration / Fetch
5	02:47:13	CREATE TABLE orders (order_id INT PRIMARY KEY, customer_id INT, order_date DATE, status V...	0 row(s) affected	0.047 sec
6	02:47:39	CREATE TABLE order_items (order_item_id INT PRIMARY KEY, order_id INT, product_id INT, quan...	0 row(s) affected	0.031 sec
7	02:48:01	INSERT INTO customers VALUES (1,'Rahul','Delhi','2025-01-10'), (2,'Aman','Mumbai','2025-02-15'), (3,'Sara','Ba...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.032 sec
8	02:48:16	INSERT INTO products VALUES (101,'Laptop','Electronics',55000), (102,'Phone','Electronics',25000), (103,'Sh...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
9	02:48:29	INSERT INTO orders VALUES (1001,1,'2025-04-01','Completed'), (1002,2,'2025-04-05','Completed'), (1003,1,2...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
10	02:48:45	INSERT INTO order_items VALUES (1,1001,101,1), (2,1002,102,2), (3,1003,103,1)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec

SQL INSERT Queries:-

```

37 );
38
39 • INSERT INTO customers VALUES
40 (1,'Rahul','Delhi','2025-01-10'),
41 (2,'Aman','Mumbai','2025-02-15'),
42 (3,'Sara','Bangalore','2025-03-01');
43
44
45 • INSERT INTO products VALUES
46 (101,'Laptop','Electronics',55000),
47 (102,'Phone','Electronics',25000),
48 (103,'Shoes','Fashion',3000);
49
50 • INSERT INTO orders VALUES
51 (1001,1,'2025-04-01','Completed'),
52 (1002,2,'2025-04-05','Completed'),
53 (1003,1,'2025-04-10','Cancelled');
54
55 • INSERT INTO order_items VALUES
56 (1,1001,101,1)

```

Output

#	Time	Action	Message
5	02:47:13	CREATE TABLE orders (order_id INT PRIMARY KEY, customer_id INT, order_date DATE, status V...	0 row(s) affected
6	02:47:39	CREATE TABLE order_items (order_item_id INT PRIMARY KEY, order_id INT, product_id INT, quan...	0 row(s) affected
7	02:48:01	INSERT INTO customers VALUES (1,'Rahul','Delhi','2025-01-10'), (2,'Aman','Mumbai','2025-02-15'), (3,'Sara','Ba...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

Total revenue :-

```

59
60 • SELECT
61     SUM(p.price * oi.quantity) AS total_revenue
62 FROM order_items oi
63 JOIN products p
64 ON oi.product_id = p.product_id;
65

```

Result Grid

total_revenue
108000.00

Top Selling Products:-

```
60
61 • SELECT
62     p.product_name,
63     SUM(oi.quantity) AS total_quantity
64 FROM products p
65 JOIN order_items oi
66 ON p.product_id = oi.product_id
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

product_name	total_quantity
Phone	2
Laptop	1
Shoes	1

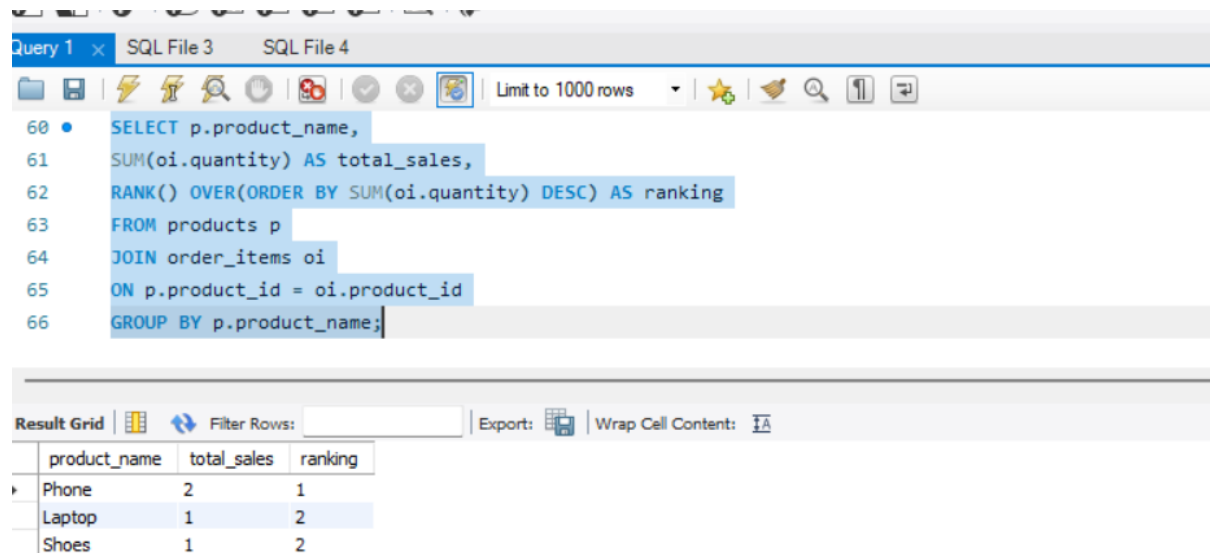
Top Customers by Revenue:-

```
60 • SELECT c.customer_name, SUM(p.price * oi.quantity) AS total_spent
61 FROM customers c
62 JOIN orders o ON c.customer_id = o.customer_id
63 JOIN order_items oi ON o.order_id = oi.order_id
64 JOIN products p ON oi.product_id = p.product_id
65 GROUP BY c.customer_name
66 ORDER BY total_spent DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

customer_name	total_spent
Rahul	58000.00
Aman	50000.00

Product Sales Ranking using Window Function:-



The screenshot shows a SQL IDE interface. The top toolbar includes icons for file operations, execution, and a 'Limit to 1000 rows' dropdown. The query editor contains the following SQL code:

```
60 • SELECT p.product_name,  
61 SUM(oi.quantity) AS total_sales,  
62 RANK() OVER(ORDER BY SUM(oi.quantity) DESC) AS ranking  
63 FROM products p  
64 JOIN order_items oi  
65 ON p.product_id = oi.product_id  
66 GROUP BY p.product_name;
```

Below the query editor is the 'Result Grid' section, which includes a 'Filter Rows' input and an 'Export' button. The results are displayed in a table with the following data:

product_name	total_sales	ranking
Phone	2	1
Laptop	1	2
Shoes	1	2

Conclusion:-

Conclusion Paragraph

This project demonstrates the use of SQL for analyzing sales data, customer behavior, and product performance. Various SQL concepts such as JOIN, GROUP BY, aggregate functions, and window functions were used to generate business insights. This project helped in improving practical knowledge of SQL and data analytics.

Skills Learned

- SQL Queries
- Table Creation
- Data Insertion
- JOIN Operations
- Aggregate Functions
- Window Functions
- Data Analysis

Thank You